

Now You See It, Now You Don't: The Age-Related Positivity Effect to Faces Disappears in Naturalistic Settings

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There is an age-related positivity effect in attention to emotional faces. However, all of these studies have relied on computer tasks where people are directed to look at faces on a screen. The primary aim of this study was to test whether the age-related positivity effect to emotional faces emerges under more naturalistic settings. The secondary aim was to test whether an own-age bias exists in attention to emotional faces and whether task ecological validity moderates any observed effect. Younger and older adults completed a naturalistic positivity effect task where they sat in a waiting room with emotional faces on the walls while their eye-gaze behavior was monitored with a mobile eye-tracker. They also completed a computer-based task that involved viewing pairs of emotional faces on a screen while wearing a mobile eye-tracker. As predicted, a positivity effect emerged in the computer-based task where older adults looked less at negative faces compared to younger adults, but no age-related positivity bias emerged in the naturalistic task. In addition, younger adults showed an own-age bias in attention to faces, and this was strongest in the naturalistic task. There was no evidence of an own-age bias in older adults for either task. These findings highlight the importance of considering ecological validity in studies of attention and emotional aging. Implications and future directions are discussed.

Keywords: the positivity effect, emotion, eye-tracking, ecological validity, own-age bias

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Faces are a highly salient social cue that reliably capture our attention (McKone & Crookes, 2007). The human brain has evolved specialized regions for processing faces (Kanwisher et al., 1999), and infants as young as 3 months old show a preference for faces compared to nonface objects (Frank et al., 2009). Facial expressions are fundamental to human communication, providing key nonverbal cues that convey someone's feelings and intentions. Therefore, attending to facial expressions has important implications for social connectedness and broader well-being (Hein et al., 2024). Several studies have now shown that there are differences in how younger and older adults allocate their attention to positive and negative emotional faces (for reviews, see Kim & Barber, 2022; Reed et al., 2014). Older adults typically attend relatively less to negative faces

and more to positive faces compared to younger adults—a phenomenon known as the age-related positivity effect (Reed & Carstensen, 2012). The age-related positivity effect is often interpreted through the lens of socioemotional selectivity theory (SST; Carstensen et al., 1999). SST proposes that as people get older, they become aware that their time left to live is limited, and as a result, emotion regulation goals are prioritized, and less emphasis is placed on knowledge-seeking goals. Goal priorities are in turn thought to direct cognitive processing, such that with increasing age, people show a preference to attend more to positive and/or less to negative information in their cognitive processing.

The age-related positivity effect has been documented across a number of cognitive domains, including attention, memory, and

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decision making (for reviews, see Kim & Barber, 2022; Reed et al., 2014). While this effect has been shown to be robust (Reed et al., 2014), inconsistencies have been reported in the literature (e.g., Fung et al., 2008; Rolison, 2019). Indeed, a meta-analytic review found that information processing constraints are an important moderator of the positivity effect (Reed et al., 2014). Specifically, unconstrained tasks where participants simply view the stimuli in a naturalistic way tend to produce stronger positivity effects than constrained tasks that require participants to make judgments about the stimuli while viewing them, demonstrating that methodological features critically influence findings on the age-related positivity effect.

Visual attention to emotional faces is one domain where the positivity effect is commonly studied. Several eye-tracking studies have now shown that older adults exhibit a preference to look less at negative and/or more at positive faces relative to their younger counterparts (Erbe et al., 2020; Isaacowitz et al., 2006a, 2006b; Knight et al., 2007). Isaacowitz et al. (2006a) were the first to demonstrate this using eye-tracking methodology. In their study, younger and older adults viewed emotional-neutral face image pairs on a computer screen while their gaze behavior was monitored with an eye-tracker. They found that older adults showed a preference to look toward happy faces and to look away from angry faces, whereas younger adults only showed a preference to look toward fearful faces. This positivity effect in attention to faces has since been replicated in subsequent research (Isaacowitz et al., 2006b; Knight et al., 2007, although see Fung et al., 2008, who failed to replicate this effect in a Chinese sample). However, a key limitation of these studies is that they used traditional viewing paradigms that involved participants passively viewing stimuli on a computer screen. While this approach involves unconstrained processing and can lead to positivity effects (Reed et al., 2014), it lacks ecological validity. Ecological validity has long been acknowledged to be an important consideration in the broader emotional aging literature (Henry et al., 2023), with some suggesting that age-related positivity effects may be situation-specific and potentially overwritten in laboratory contexts that mimic real-life situations (Kunzmann & Isaacowitz, 2017). Indeed, high ecological validity may represent a critical boundary condition for the positivity effect. While SST posits that older adults prioritize emotional goals, leading to an attentional preference for positive emotional information, the vast majority of this evidence is drawn from highly controlled lab studies that explicitly direct participants' attention (often toward a computer screen) with minimal distractions. However, real-life environments are much more complex, and attention is often divided, which may limit the extent to which older adults can, or are motivated to, engage in selective attention to service emotion regulation goals. If positivity effects are weakened or absent in naturalistic contexts, SST and its predictions may not generalize to all contexts. This therefore raises an important question: Does the positivity effect in visual attention to emotional faces still emerge when more ecologically valid paradigms are used?

While no study to date has directly tested this possibility, three studies have used more ecologically valid designs with nonface stimuli and have failed to find an age-related positivity effect (DiGirolamo et al., 2023; Isaacowitz et al., 2015, 2018). DiGirolamo et al. (2023) measured visual attention (using stationary eye-tracking) to emotional film clips on a computer in the lab and on a computer in a more naturalistic environment—the participants' own

home. They found that both younger and older adults fixated more on positive compared to negative information in the lab, but the reverse pattern emerged in the home such that these same participants looked more at negative information. While this suggests that positivity effects are attenuated in naturalistic settings, this conclusion is tentative as there was also no clear evidence of an age-related difference in positivity effect in the lab. An additional issue is that both the lab and at-home tasks were completed on a computer, which limits ecological validity; it is not clear how younger and older adults may allocate their attention in real-world environments.

Using a different approach, Isaacowitz et al. (2015) used mobile eye-tracking to test whether a positivity effect emerged when participants were free to self-select the emotional content they engaged with. They created an "affective environment" that included three computer screens, as well as wall hangings, that presented news media (i.e., news reports) that varied in emotional valence (i.e., positive, negative, and neutral). They fitted younger and older participants with eye-tracking glasses and left them alone in the affective environment for 15 min. Participants were told that they could engage with any of the media content in the room for as little or as long as they wished. They found that overall participants looked more at positive and neutral content compared to negative content, but this effect was not influenced by age. These findings were later replicated by Isaacowitz et al. (2018) who again showed no age-related differences in attention to the different valence categories of emotion using a similar paradigm. One limitation in interpreting these results is that they did not assess the age-related positivity effect using a traditional computer viewing paradigm for comparison. Individual difference factors, such as mood and future time perspective, can influence positivity effects (Wadlinger & Isaacowitz, 2006), as well as the magnitude of the age difference between the younger and older adult sample (Reed et al., 2014). It is therefore important to establish, within the same sample, whether positivity effects seen in traditional viewing tasks also emerge with more ecologically valid tasks. This direct comparison is necessary to provide the most accurate evaluation of whether the age-related positivity effect generalizes to more naturalistic contexts. Therefore, the primary aim of this study was to test whether a positivity effect in attention to emotional faces would be evident in a traditional viewing paradigm and, critically, whether this same positivity effect would also emerge using a more ecologically valid paradigm.

Is There an Own-Age Bias in Emotion Processing?

As a secondary aim, this study also examined how younger and older adults' attention to faces might vary based upon the age of the face being observed. Faces vary considerably with respect to their features, but one particularly influential feature may be the age of the face being observed. An *own-age bias* refers to preferential processing of faces that are close to one's own age compared to another age (Rhodes & Anastasi, 2012). There are different theoretical reasons to expect an own-age bias in face processing. For instance, the perceptual expertise account suggests that the own-age bias emerges because we have spent more time with people our own age relative to other age groups, which leads to greater expertise in processing facial features from our own age group (Rhodes & Anastasi, 2012).

However, evidence for an own-age bias in attention to faces is mixed (Ebner et al., 2011; Firestone et al., 2007; Y. He et al., 2011; Strickland-Hughes et al., 2020). For example, Ebner et al. (2011)

presented participants with images of younger and older faces depicting different emotional expressions (i.e., happy, angry, sad, fearful, and disgusted). Participants were simply asked to view the faces as if they were watching television. They found an own-age bias for younger and older adults, whereby older adults gazed longer at older faces, and younger adults gazed longer at younger faces. This pattern of results was replicated by Y. He et al. (2011). However, other work has failed to find an own-age bias. For instance, Firestone et al. (2007) presented participants with younger and older famous faces and found that both groups looked more at younger faces compared to older faces. Strickland-Hughes et al. (2020) asked participants to complete a face-name association task and measured visual attention to emotional faces during both encoding and recall. They found younger and older adults showed a similar allocation of visual attention to own-age and other-age faces. Given these mixed findings, further work is needed to clarify whether an own-age bias exists in response to emotional faces.

The categorization-individuation model (CIM) argues that people engage in one of two perceptual processes when they encounter others: categorization and individuation (Hugenberg et al., 2010). Initially, people engage in spontaneous categorization, focusing on facial features that are diagnostic of broad social categories, such as age, race, and sex. This allows for quick social categorization of others. Attention can then be reallocated toward identity-diagnostic features, allowing finer discrimination between individuals within the same social category. However, engaging in individuation is theorized to be more likely to occur when the target is categorized as an ingroup member, and this may lead people to look longer at own-age faces as they seek to differentiate between members within their own group. In contrast, people may spend less time looking at other-age faces as these are processed more categorically, with less attention paid to the individuating features. Importantly though, Hugenberg et al. (2010) argued that engagement in this latter identity processing is dependent on motivation, whereby individuals will only do so if they view the target as important or meaningful. Given that older adults are motivated to attend to positive information, they may engage in identity processing (and therefore longer looking) for positive faces regardless of face age. As a result, for older adults, the own-age bias in attention may not emerge for positive faces, while still occurring for negative faces.

While the CIM leads to the prediction that older adults' own-age bias may vary based on face valence, it is also important to acknowledge that emotional expressions portrayed by older faces are often more ambiguous. As a result of age-related changes in physical features (e.g., facial measures, wrinkles), emotions are often perceived less accurately in older compared to younger faces (Ebner & Johnson, 2009; Ebner et al., 2011). This means that the emotional content of the faces may be processed less accurately in older faces (see Fernandes et al., 2023), and consequently, if emotion does influence the own-age bias, any effect may be weaker for older faces.

Another important question that remains in the own-age bias and visual attention literature is whether the effect emerges in more naturalistic settings. All of the own-age bias studies to date have been conducted using traditional viewing tasks where younger and older adults are instructed to look at a computer screen. Given that motivation is implicated in theories of the own-age bias (e.g., CIM), and in theories of emotional aging more broadly, it is possible that the own-age bias is larger under more naturalistic conditions. This is

because naturalistic tasks are less controlled and give participants more freedom in how they behave and respond, which may mean that any effect motivation may be amplified in the naturalistic task. The second aim of this study was therefore to test whether there is an own-age bias in visual attention to faces and, critically, whether this differs as a function of emotional valence and task ecological validity.

Hypotheses

The positivity effect is defined as an Age $\times$ Valence interaction (Barber & Kim, 2022; Reed & Carstensen, 2012). We tested for a positivity effect in attention to emotional faces in a traditional viewing paradigm and in a more naturalistic viewing task. We expected to find that the magnitude of the positivity effect varied between these two tasks, whereby the positivity effect emerged more strongly in the traditional task but was attenuated (or absent) in the naturalistic task.

As noted above, the evidence is mixed with respect to own-age biases in attention to emotional faces (Ebner et al., 2011; Firestone et al., 2007; Y. He et al., 2011; Strickland-Hughes et al., 2020). It was also unclear whether, or how, task ecological validity may affect the own-age bias; therefore, we kept this second question exploratory and did not make any *a priori* predictions.

Method

Transparency and Openness

Our study design and hypotheses were preregistered on the Open Science Framework (see Ward et al., 2025) and can be accessed via the following link: <https://osf.io/vrkhp/>. Below, we report how our sample size was determined, the key exclusion criteria, and the measures used for this study. We provided details in the author note regarding how to access data and analytic code used for this study.

Participants

An *a priori* power analysis revealed that a minimum of 82 participants (41 per age group) were needed to detect a moderate-sized ($f = .24$) within-between interaction (i.e., Emotion $\times$ Age Group; see Isaacowitz et al., 2006b) with 80% power. We initially recruited 62 younger adults and 53 older adults, but we had hardware problems with the mobile eye-tracker and were unable to collect data for 13 younger and 4 older adults. The final sample therefore included 49 young adults (17–40 years, 36 females, 65% Caucasian) and 49 older adults (65–85 years, 35 females, 100% Caucasian). Demographic characteristics for the sample are reported in Table 1. Older adults reported more full-time education than younger adults. The groups did not differ in self-reported health, but older adults reported less anxiety compared to younger adults (indexed via the Hospital Anxiety and Depression Scale; Zigmond & Snaith, 1983). No age differences were found in depressive symptoms. Consistent with prior aging studies (Grainger et al., 2017; Knight et al., 2007), older adults demonstrated greater crystallized intelligence relative to young (indexed via the National Adult Reading Test; Nelson, 1982).

The younger adults were recruited through the University of Queensland 1st-year participation scheme and received two course credits for participating. Older adults were recruited from the local community via our older adult research registry as well as via

Table 1
Demographic Information for Younger and Older Adults

Measure	Young adults			Older adults			Statistics		
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>	<i>t</i>	<i>df</i>	<i>p</i>
Age	49	19.94	4.29	49	72.29	4.96	55.90	96	<.001
Education	49	13.82	1.62	49	16.49	5.20	3.44	96	<.001
Health	49	3.51	0.92	49	3.43	0.98	0.43	96	.671
HADS A	47	8.89	3.65	49	6.16	3.41	3.79	94	<.001
HADS D	45	4.04	3.08	49	5.06	2.93	1.64	92	.105
NART	48	117.46	3.64	47	122.15	3.74	6.60	93	<.001

Note. Health refers to self-rated health, which was measured with the following question: “How would you rate your overall health?” Responses were made on a 5-point scale, with higher scores indicating better health. HADS A = the Anxiety subscale of the Hospital Anxiety and Depression Scale; HADS D = the Depression subscale of the Hospital Anxiety and Depression Scale; NART = National Adult Reading Test, which is a measure of estimated IQ.

Facebook advertisements, and they received \$40 in gift cards for their participation. Prior to participation, older adults were screened for cognitive impairment using the Mini-Addenbrooke’s Cognitive Examination (Hsieh et al., 2015), and all scored above the cutoff of 21 (out of 30) to participate in this study ($M = 28.36$, $SD = 2.03$, range = 22–30). Participants completed a screening questionnaire prior to taking part in the study and were not eligible to participate if they self-disclosed having a neurodevelopmental, neurodegenerative, or psychiatric condition. Participants were also required to have English as a first language and normal or corrected-to-normal vision. This project was conducted in Brisbane at an urban university, and all data were collected from April to August 2024. This project was approved by the Human Research Ethics Committee at the University of Queensland.

Measures

Traditional Viewing Paradigm

We used a similar design to Isaacowitz et al. (2006b). The emotional face stimuli were taken from the FACES database (Ebner et al., 2010). Participants viewed pairs of images that contained a neutral face paired with either a happy (i.e., positive) or an angry (i.e., negative) face. Younger and older, male and female, protagonists were used, and the same protagonist was used for both images in each face pair. Half the image pairs presented the emotional expression on the left side of the screen, and the other half presented it on the right. As per Isaacowitz et al., participants were simply required to view the images on the screen while wearing a pair of eye-tracking glasses (more details regarding the eye-tracking methodology are detailed below). Tobii Lab Pro was used to program the experiment. Each trial was preceded by a central fixation cross, and the image pair was presented for 4 s. Forty-eight image pairs were presented in total, and the order of presentation was randomized across participants. The task took approximately 4 min to complete.

Naturalistic Positivity Effect Task

For this task, we aimed to mimic a real-life situation—a waiting room—in the laboratory. This approach allowed us to enhance

ecological validity while also maintaining some degree of experimental control. We chose a waiting room scenario because this is a commonly experienced situation for both younger and older adults, and it provided an ideal opportunity to measure *spontaneous* visual attention to emotional information. Our mock waiting room was an adjoining lab that had workstations and chairs and 24 images of happy (i.e., positive), angry (i.e., negative), and neutral faces hanging on the wall. Consistent with the traditional viewing task, the stimuli were drawn from the FACES database (Ebner et al., 2010) and included older and younger, male and female, protagonists. A new set of images was selected for this task so that participants never saw the same image twice across the two tasks. The images were printed in A4 size and laminated before attaching to the wall. The arrangement of the images on the wall was randomized across participants. As will be explained in the Procedure section, participants were simply asked to wait in this waiting room while wearing eye-tracking glasses.

Apparatus

Tobii Glasses 3

Tobii Glasses 3 is a mobile eye-tracker that was used to measure participants’ eye movements in both positivity effect tasks. Although prior work examining the positivity effect in traditional tasks has used screen-based eye-trackers (e.g., Isaacowitz et al., 2006a, 2006b), we decided to use mobile eye-tracking in both tasks to avoid the glasses being a measurement confound. Specifically, although the glasses are very discreet, some participants may have been more aware that their attention was being monitored when wearing the glasses and, as such, could have behaved differently (Risko & Kingstone, 2011). Therefore, participants wore the glasses in both tasks to keep the eye-tracking measurement type consistent. The Tobii Glasses record eye movements at a rate of 50 Hz and provide highly accurate and reliable eye-tracking data with minimal data loss (Onkhar et al., 2024; Shoenfelt et al., 2025). We planned to exclude any participants who were tracked for less than 70% of the time, but no participants were excluded based on this criterion. If the participant wore corrective lenses, the Tobii Glasses were fitted with Tobii’s snap-on corrective lenses at the appropriate prescription. If the participant did not know their prescription, they completed a Snellen eye test to determine their required prescription. We used a temporal threshold of 100 ms for fixations and a visual angle of 1°. Fixation time refers to the total amount of time that a participant’s gaze remains in fixation for a given time window. It is more sensitive in detecting attentional biases compared to fixation count; therefore, this was the key metric extracted for this study. As detailed below, we created a percent fixation time score (see Grainger & Henry, 2023; Isaacowitz et al., 2006a, 2006b) to account for individual differences in overall fixation time. Percent fixation time was calculated by dividing the fixation time in an area of interest (AOI) by the total fixation time for that trial or time window. All raw eye data were analyzed using Tobii Lab Pro.

Eye-Tracking AOIs

Tobii Lab Pro was used to create AOIs for the traditional and naturalistic tasks. For the traditional task, an AOI was drawn around each image (i.e., emotional, neutral) in the pair (i.e., two AOIs per

trial), and total fixation time gazing at each image was quantified. For the naturalistic task, a separate AOI was created for each of the 24 images of faces on the wall (24 AOIs in total), allowing % fixation time to each image to be quantified. We created a mean composite score for fixation time for each emotion category (i.e., happy, angry, neutral) separately for each task (i.e., traditional, naturalistic). For the traditional task, we calculated the % fixation time score for each AOI by dividing the fixation time for each AOI by the total fixation time for the trial. This was then multiplied by 100 to create a percentage score, and these percentage scores were then averaged to create a mean % fixation time score for each emotion category (i.e., positive, negative, and neutral). For the naturalistic task, all stimuli were presented in a single array for the 2-min time window; therefore, we summed the fixation time for the eight images in each emotional category (i.e., to create a total fixation time score for each emotion) and then divided these by the total fixation time for the 2-min trial. These were then multiplied by 100 to create percentage scores.

Procedure

All testing sessions took place in a laboratory at the University of Queensland. Before beginning the session, the participants were given an information sheet and were asked to sign a consent form. Next, they were asked to place their phones in their bags or in a dedicated box while on silent. This was done to ensure participants were not able to access their phones during the naturalistic task. Next, participants completed the National Adult Reading Test and demographics survey. Once participants completed these initial measures, they were fitted with the Tobii Glasses 3 and completed the calibration procedure. To calibrate the glasses, participants were asked to look at a Tobii calibration card that was fixed to a wall in the lab while standing 0.5 m away. Once calibration was completed, participants were asked to look at three places in the room to ensure the eye-tracker was accurately capturing their gaze (as per Tobii's recommendations). Next, the participant completed the eye-tracking tasks, which were presented in a counterbalanced order across participants. For the naturalistic task, participants were taken into an adjoining room with an "Emotional Face Space" sign on the door and were asked to take a seat and wait "while the experimenter set up the next task from the computer in the other room." Critically, the wall directly in front of the participants had the 24 emotional face images attached. The experimenter mentioned when they entered the room that this is an emotion lab, hence the emotional faces on the wall, but made no further mention of the faces and, critically, provided no explicit instruction to attend to the faces. The participant was left in the waiting room for 2 min. After 2 min, the experimenter returned to the room and explained that there were "technical errors" and that they would just move on to the next task in the other room. For the traditional viewing task, we used a similar procedure and instructions to Isaacowitz et al. (2006b). Participants were seated in front of a computer monitor and were told, "This is an emotional processing task, the computer is going to show you a series of images of people showing different emotions, prior to each image you will see a fixation cross, please focus your attention to the cross and once the cross has disappeared please view the faces as you would if you were watching tv." They then passively viewed the stimuli until all trials had been presented.

Once these two tasks were completed, the glasses were removed, and participants completed tasks for another unrelated experiment as well as other self-report questionnaires. At the completion of these tasks, participants were debriefed and compensated for their time.

Data Analysis

All analyses were conducted in SPSS (Version 29) and JASP (0.19.3). In our original preregistration, we planned to use fixation ratio scores for the traditional task (as per Isaacowitz et al., 2006a, 2006b) and raw scores for the naturalistic task because a fixation ratio score was not possible due to the stimuli not being presented in emotion-neutral pairs. However, we decided that it would be more appropriate to analyze % fixation time in both tasks to increase comparability.¹ We conducted 2 (participant age group: younger, older) $\times$ 3 (emotion: positive, negative, neutral) $\times$ 2 (face age: young, old) mixed frequentist and Bayesian ANOVAs, separately for each task (traditional, naturalistic).² See the [Supplemental Material](#) for these analyses conducted with mean fixation time (ms). Any interactions were followed up using paired-samples and independent-groups *t* tests as appropriate. Cohen's *d* and η_p^2 were used to quantify effect sizes. α was set at $p < .05$. For the Bayesian analyses, we used the default Cauchy prior ($r = 0.50$) and calculated inclusion Bayes factors (BF_{incl}) for each main effect and interaction using model averaging across matched models. Bayes factors are presented in Table 2. We followed recommendations by van Doorn et al. (2021) to interpret the strength of Bayesian evidence.

Results

Traditional Task

In the ANOVA examining % fixation times during the traditional task, there was a main effect of emotion, $F(2, 188) = 10.38$, $p < .001$, $\eta_p^2 = .10$, but this was qualified by a two-way interaction between age group and emotion, $F(2, 188) = 5.96$, $p = .003$, $\eta_p^2 = .06$. The corresponding BF_{incl} was 3.060, providing moderate evidence in favor of the interaction effect. To better understand this interaction, we compared % fixation time across emotion categories separately for each age group. These analyses revealed that older adults looked less at negative faces compared to neutral faces, $t(48) = 2.21$, $p = .032$, $d = 0.54$, and positive faces, $t(47) = 2.50$, $p = .016$, $d = 0.62$. For younger adults, the opposite pattern emerged, whereby younger adults looked more at negative faces relative to neutral faces, $t(48) = 3.35$, $p = .002$, $d = 0.67$. They also looked less at neutral faces, $t(48) = 5.61$, $p < .001$, $d = 1.05$, and negative faces, $t(48) = 2.29$, $p = .002$, $d = 0.67$, relative to positive faces. These results are presented visually in Figure 1A. The only other significant effect was a main effect of target face age, $F(1, 94) = 4.96$, $p = .028$, $\eta_p^2 = .05$, indicating that young faces were looked at longer than older faces, $t(96) = 2.04$, $p = .044$, $d = 0.21$ (see Figure 2A). However, the Bayesian analyses provided moderate evidence against an effect of target face age ($BF_{incl} = 0.214$). The interaction

¹ We analyzed the data from the traditional task using fixation ratio scores, and the findings were consistent with the findings generated with the % fixation time scores.

² There were no significant correlations between % fixation time and anxiety for either age group ($r_s < .15$, $p_s > .197$); therefore, this was not included as a covariate in the analyses.

Table 2
BF_{incl} for All Main Effects and Interactions, Separately by Task

Effect	BF _{incl}	Interpretation
Traditional task		
Emotion	238.29	Strong evidence in favor of an effect
Target face age	0.214	Moderate evidence against an effect
Emotion × Target Face Age	0.082	Strong evidence against an effect
Age group	0.632	Inconclusive
Emotion × Age Group	3.060	Moderate evidence in favor of an effect
Face Age × Age Group	0.078	Strong evidence against an effect
Emotion × Face Age × Age Group	0.005	Strong evidence against an effect
Naturalistic task		
Emotion	0.056	Strong evidence against an effect
Target face age	269.59	Strong evidence for an effect
Emotion × Target Face Age	0.201	Moderate evidence against an effect
Age group	0.239	Moderate evidence against an effect
Emotion × Age Group	0.028	Strong evidence against an effect
Face Age × Age Group	0.960	Inconclusive
Emotion × Face Age × Age Group	0.068	Strong evidence against an effect

Note. BF_{incl} = inclusion Bayes factors.

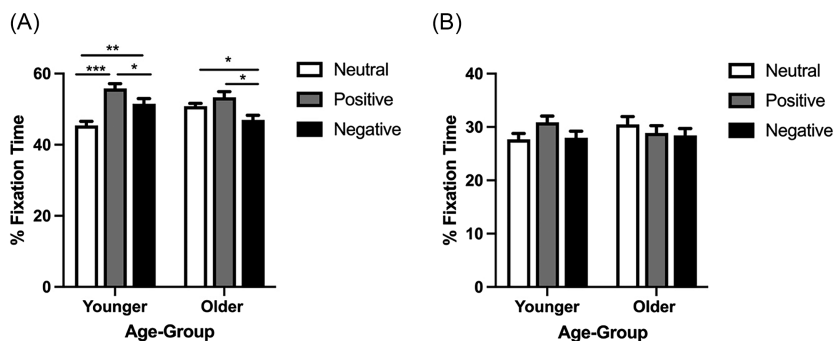
between face age and emotion was not significant, $F(2, 188) = 1.03$, $p = .360$, $\eta_p^2 = .01$, nor was the interaction between face age and age group, $F(2, 188) = 0.55$, $p = .461$, $\eta_p^2 = .01$. The three-way interaction involving emotion, face age, and age group was also not significant, $F(2, 188) = 0.21$, $p = .811$, $\eta_p^2 < .01$.

Naturalistic Task

In the ANOVA examining % fixation times during the naturalistic task, there was no main effect of emotion, $F(2, 188) = 0.68$, $p = .510$, $\eta_p^2 = .01$, and no interaction between emotion and age group, $F(2, 188) = 1.37$, $p = .257$, $\eta_p^2 = .01$. As can be seen in Table 2, BF_{incl} = 0.028 for the interaction between emotion and age group, providing strong evidence for no interaction effect. These results are presented visually in Figure 1B. However, there was a main effect of target face age, $F(1, 95) = 22.04$, $p < .001$, $\eta_p^2 = .19$, but this was superseded by an interaction between target face age and participant age group, $F(1, 95) = 8.20$, $p = .005$, $\eta_p^2 = .08$. This was followed up by testing the effect of target face age separately for each age group. This revealed a

significant own-age bias for younger adults, with greater time spent looking at younger relative to older faces, $t(47) = 5.12$, $p < .001$, $d = 1.12$. There was no effect of target face age for older adults, $t(48) = 1.35$, $p = .182$, $d = 0.29$. See Figure 2B. It is worth noting that while the Bayesian evidence for an effect of task age was very strong (BF_{incl} = 269.592), the evidence for the interaction between target face age and age group was inconclusive (BF_{incl} = 0.960). The interaction between face age and emotion was significant, $F(2, 190) = 3.49$, $p = .033$, $\eta_p^2 = .04$, although Bayesian analyses showed moderate evidence against an effect (BF_{incl} = 0.201). This was followed up by looking at the effect of target face age for each emotion. For positive and neutral faces, younger faces were looked at longer than older faces, $t(96) = 3.65$, $p < .001$, $d = 0.51$; $t(96) = 4.11$, $p < .001$, $d = 0.56$, respectively. There was no effect of target face age for negative faces, $t(96) = 0.69$, $p = .491$, $d = 0.10$. The three-way interaction with age group, emotion, and target face age was not significant, $F(2, 190) = 2.29$, $p = .104$, $\eta_p^2 = .02$, and the Bayesian analyses provided strong evidence against an interaction effect (BF_{incl} = 0.068).

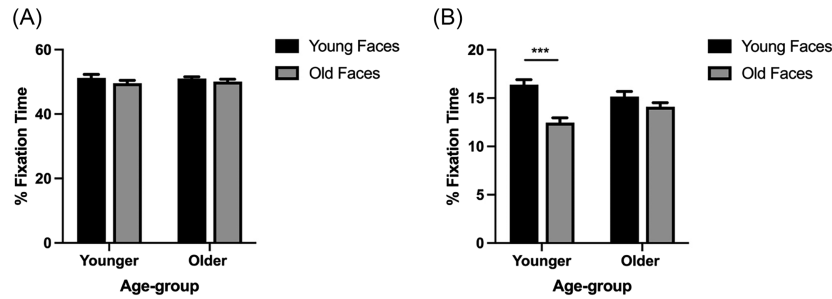
Figure 1
Fixation Time (%) to Positive, Negative, and Neutral Faces, Separately for Younger and Older Adults in the (A) Traditional Task and (B) Naturalistic Task



* $p < .05$. ** $p < .01$. *** $p < .001$.

Figure 2

Fixation Time (%) to Younger and Older Faces Separately for Younger and Older Adults in the (A) Traditional and (B) Naturalistic Tasks



*** $p < .001$.

Discussion

Several studies have shown that older adults display a positivity effect in their visual attention to emotional faces (Isaacowitz et al., 2006a, 2006b; Knight et al., 2007). However, all of these studies have used traditional viewing paradigms completed on a computer, which lack ecological validity. Therefore, the primary aim of this study was to test whether a positivity effect emerges in response to emotional faces when a more ecologically valid paradigm is used. To test this, we used mobile eye-tracking to measure younger and older adults' gaze behavior to emotional faces while they waited in a mock waiting room with pictures of emotional faces on the walls. Importantly, we also asked participants to complete a traditional viewing paradigm to match previously reported effects (Isaacowitz et al., 2006a, 2006b) and to compare with the naturalistic viewing task.

In line with our preregistered predictions, we found an age-related positivity effect on the traditional viewing task, whereby older adults looked less at negative images and more at neutral images relative to younger adults. The Bayesian analyses indicated that there was moderate evidence for this interaction effect. This aligns with prior work that has identified a positivity effect in visual attention (Isaacowitz et al., 2006a, 2006b; Knight et al., 2007). Critically though, there was no evidence for a positivity effect on the naturalistic viewing task, with no differences in younger and older adults' allocation of gaze across the three emotion categories (i.e., positive, negative, and neutral facial expressions). This absence of an interaction effect was supported by Bayesian analyses, which revealed strong evidence against any interaction effect. These findings show for the first time that the positivity effect in visual attention to faces is less robust when a more ecologically valid task is used, adding to a growing literature that suggests ecological validity is an important consideration when studying the age-related positivity effect (Alea et al., 2004; Devarayapuram Ramakrishnan et al., 2024; DiGirolamo et al., 2023; Isaacowitz et al., 2015, 2018; Rolison, 2019). These findings have important implications for theories of emotional aging such as SST, which posit that older adults are motivated to prioritize emotional goals and may do so by showing a preference away from negative emotional information. Our results showed that older adults do indeed show this preference, but only when they are told to direct their attention to the emotional information on a computer screen, without any other competing

distractors. This suggests that the positivity effect in attention may only exist in highly controlled situations, such as research laboratories, and may not generalize to real-world environments, which are inherently more complex. Prior work has shown that in traditional tasks, the positivity effect is stronger during unconstrained tasks, where participants are instructed to freely view images (Reed et al., 2014). On the surface, our findings seem to contradict this conclusion. This is because participants were able to freely view the images on the walls in the naturalistic task, with no other task demands. Yet, unlike the traditional task, we found no positivity effect for older adults. One key difference between the traditional and naturalistic tasks was the instructions. Whereas in the traditional task, participants were instructed to "view the faces as if they were watching tv," the naturalistic task included no instructions to view the faces. We chose not to instruct participants to view the faces to enhance ecological validity, because in day-to-day life we are often not directed to look at certain aspects of our environment; instead, our attention is captured spontaneously. However, it is possible that the lack of instruction to focus participants' attention on the emotional faces on the walls could have led to older adults' attention being divided and other (nonemotional) goals being prioritized. Indeed, this interpretation aligns with Williams et al. (2020), who found that a positivity effect in response inhibition only emerged when participants were explicitly instructed to focus on emotion-relevant information. While testing goal-directed attention to emotion was not the focus of our naturalistic task—which aimed to examine spontaneous gazing to emotion—it is possible that a different pattern of results may have emerged if participants were given an explicit goal or task that required them to focus on the faces in the naturalistic task (e.g., being told these were faces they would need to remember in the future). Future experimental work is therefore needed to test precisely how manipulating goals or task instructions may modulate younger and older adults' attention under more naturalistic settings. However, it is worth noting that Isaacowitz et al. (2015), who also used mobile eye-tracking, explicitly instructed participants to interact with the emotional information in their "affective environment," making the emotional information salient, but they also failed to find a positivity effect in older adults' attention. They concluded that the novel environment may have led to a greater focus on information-seeking goals for both groups (and less focus on emotion regulation goals), which may have meant older adults did not actively prioritize positive information and avoid negative

information. This interpretation also aligns with our findings—participants were taken into a new room for the naturalistic task, which may have led to them adopting a more information-focused mindset, placing less emphasis on emotion regulation goals.

Another methodological difference between the traditional and naturalistic tasks in this study was the presentation of the facial stimuli. Whereas the traditional task presented the stimuli in a series of emotional-neutral face pairs for 4 s per trial, the naturalistic task presented all emotional faces (i.e., positive, negative, and neutral) in a single array. Knight et al. (2007) identified an age-related positivity effect with neutral-emotional face pairs but not emotion(positive)-emotion(negative) face pairs, which suggests that the positivity effect may only emerge when an emotional stimulus is directly contrasted with a neutral stimulus. Future studies should aim to develop naturalistic paradigms that allow for pairs of emotional and neutral stimuli to be viewed for brief periods, more closely matching the presentation style of traditional tasks. A mobile eye-tracking paradigm that involves walking through a real-world environment (see De Lillo et al., 2021) where emotional-neutral information is scattered throughout could be a promising option for future studies. While it is not possible to conclude from our data precisely why participants did not show a positivity effect in the naturalistic task, the key point to take away is that the positivity effect does not appear to be robust across task types, and future work is now needed to understand why this might be.

The second aim of this study was to test whether there is an own-age bias in younger and older adults' attention to emotional faces and, critically, to provide the first test of whether the own-age bias is moderated by emotional valence and task type. This research question was exploratory given that the literature is mixed with respect to own-age bias in visual attention (Ebner et al., 2011; Firestone et al., 2007; Y. He et al., 2011; Strickland-Hughes et al., 2020). Across both tasks, there was a significant effect of target face age, whereby younger faces were looked at more than older faces. This aligns with early work by Firestone et al. (2007), who found that younger faces were looked at longer than older faces for both younger and older participants. This tendency to look at younger faces could be driven by differences in perceptions of attractiveness for young and old faces. Prior work has shown that people look longer at attractive compared to nonattractive faces and are faster to respond to faces when they are perceived as attractive (Leder et al., 2010). Moreover, prior work has also shown that older faces are rated as less attractive than younger faces—an effect that is more pronounced in younger participants (Ebner et al., 2018; D. He et al., 2021). Critically though, the tendency to look more at younger faces was considerably larger in the naturalistic task, and there was also an interaction with emotion. This reflected the fact that for both age groups, younger faces were looked at longer than old, but only when they were displaying happy and neutral expressions. This finding aligns with neuroimaging evidence from Ebner et al. (2013), who showed that the activity in the medial prefrontal cortex and insula was stronger for own-age compared to other-age faces but only when faces were showing happy or neutral displays, suggesting that angry expression may override target face age effects. We also found an interaction between target face age and age group in the naturalistic task. Specifically, whereas younger adults looked considerably longer at young faces compared to older (i.e., indicative of an own-age bias), older adults showed no differences in gaze behavior as a function of face age. This finding is consistent with

Denkinger and Kinn (2018), who found an own-age bias in face recognition in younger but not older adults. While the own-age bias in younger adults is in line with theories such as CIM and perceptual expertise, the lack of an own-age bias in our older sample is not. However, CIM posits that engaging in individuation is motivation-dependent and may only occur if the targets are perceived to be meaningful. Given that the stimuli were strangers, it is possible that the older adults did not perceive them as meaningful, and as such, they did not engage in individuation. Prior work has shown that the meaningfulness of stimuli is particularly important for older adults in emotion studies (Fung et al., 2019; Stanley & Isaacowitz, 2015). Further work is now needed to better understand the specific task features that may be important for the own-age bias to emerge.

Strengths and Limitations

This study had many strengths, including using the same eye-tracking methodology and validated emotion stimuli across the two counterbalanced tasks, which reduced the possibility that methodological factors could explain any differences in task performance. However, while this study provided the first test of the positivity effect on emotional faces using a more ecologically valid paradigm, the task was still conducted within the confines of a testing laboratory, which limits the ecological validity of the design. Future research could build on this by developing a positivity effect paradigm that can be completed in a real-world environment, such as a local library or museum. Another limitation was that the stimuli were photographs of strangers as opposed to close others. While this was done to keep methodological factors consistent across the traditional and naturalistic tasks, it is unclear whether these findings would replicate if images of close others or familiar people were used as stimuli, as opposed to images of novel strangers (see Henry et al., 2023). Indeed, stimuli meaningfulness may be particularly important when studying the own-age bias. Future work could therefore build on this study by including more meaningful stimuli (i.e., familiar people).

Finally, while a strength of this study is that we preregistered our sample size and hypotheses, our sample size calculation was based on the moderate effect size reported in Isaacowitz et al. (2006b), who were the first to identify the age-related positivity effect in visual attention. While this sample size provided sufficient power to detect a positivity effect on our traditional task, it may not have been sufficiently powered to detect an effect in the naturalistic task if it was smaller in magnitude. Therefore, another interpretation could be that the positivity effect does indeed exist with more ecologically valid designs, but it is much smaller in magnitude. If this were the case, the practical implications of a very small effect might be relatively unimportant for everyday behavior, and it would be difficult to detect in mobile eye-tracking studies where it is challenging to collect large sample sizes. To conclude, this study tested whether the age-related positivity effect in visual attention is present when a more ecologically valid task is used. We showed that while the positivity effect emerged on the traditional viewing task, such that older adults looked less at negative compared to positive faces, there were no age differences in visual attention during the more naturalistic viewing task. These findings add to a growing literature suggesting that the positivity effect may not generalize to more ecologically valid task designs and therefore raise questions with respect to whether it even exists in the real world or if it is just an

artifact of lab-based studies. We also found an own-age bias in attention to faces for younger adults across both tasks, and this effect was particularly strong in the more naturalistic task. Together, these findings highlight the importance of considering ecological validity when measuring visual attention to emotional faces in older age, as well as the need for more naturalistic tests of the positivity effect in future studies.

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